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TCMS - Operational-VF.CONCEPT

SFRS for Zefiro-Europe

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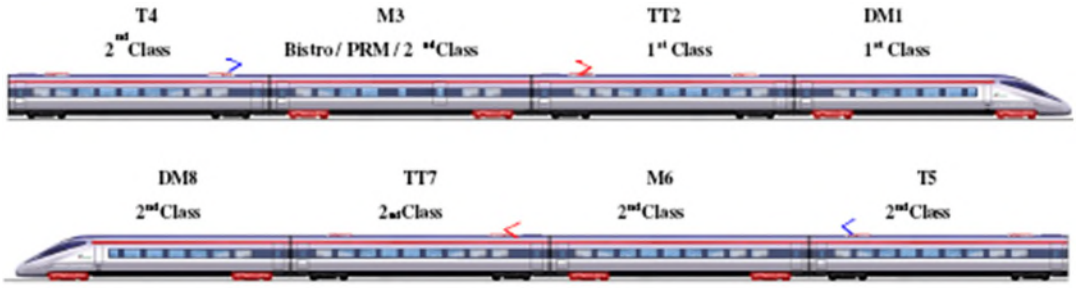
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1 Introduction

This document is a function specification for the control of the *TCMS Operational* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

BA	Bus Administrator
CD	Circuit Diagram
CCU	Central Control Unit
VCU	Vehicle Control Unit
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
ETB	Ethernet Train Backbone
ECN	Ethernet Consist Network (ring)
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
IPT	Internet Protocol - Train
MCG	Mobile Communication Gateway
NTP	Network Time Protocol
TCMS	Train Control and Management System
TWCS	Train to Wayside Communication System
MVB	Multifunctional Vehicle Bus
WTB	Wired Train Bus

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2 System Overview

The task for the TCMS system is to provide a platform for software controlled functionalities of the train including; Controllers (VCU, IDU etc.), Communication between controllers and Interface to external electrical equipment

The TCMS System identifies the train configuration and make this information available for the software controlled functions in the train.

The TCMS Operational is reading/writing signals from/to external electrical systems and makes that information available for functionality implemented in the VCU.
In addition some base functionality is includes (time synchronization, redundancy, etc).

Note: MCG not included, part of TWCS.

2.1 System Configuration

Failure Modes:

The following failure modes are defined within TCMS electronics. Each individual mode is detected by the unit and reported to the application software. The application software generates individual events for those failures. TCMS redundancy is designed to cope with any single point failure. Hence as long as one event is occurring stand alone, it does not affect the operation of the vehicle.

1. Component failure: Any TCMS electronics unit is defective. All in/outputs are de-energised. Error may be caused by power supply failure, supply connector failure, CPU failure.
2. Failure of specific input step: Applicable to MIO, the actual input gets high impedative, the inputs data is neutralised in software.
3. Failure of specific output step: Applicable to MIO, the actual output gets high impedative to neutralise the effect on the electrical circuits.
4. IDU component failure: failures on sub-components as touch screen, background light or CPU will cause a total IDU failure.
5. Cable and Contactor Socket failures are not classified as TCMS failures, they are rather depending on manufacturing problems. But as they will effect the TCMS performance they are considered.

Redundancy Concept:

The following redundancy measures have been taken to improve the system reliability:

1. Redundant MVB: the MVB is of type "dual line EMD" to cope with single point failures in the contactor sockets or in the MVB cabling. In addition two redundant MVB buses are used.
2. Redundant BA running in the CCU: this is a functional redundancy to keep full MVB performance in case of a single BA failure.
3. Redundant I/O: important digital inputs from the train wiring have been duplicated to cope with single cable or TCMS MIO component failures. Duplicated signals are e.g. the input from the cab activation key switch.
4. ECN redundancy: this redundancy is build on a ring structure. The IP ring communication performance will be kept in case a single ring switch or cable failure occurs.
5. ETB redundancy: the ETB is connected using two separate communication cables between each train switch / train repeater.
6. Redundant WTB: the WTB is connected using two separate communication cables between the TCN GWs.
7. Redundant IDU: the primary IDU is the interface to the driver, in case this fails it is possible to activate (by replacing) the back-up IDU.

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8. Redundant CCU:
   The CCU-O is redundant (warm standby), if one CCU-O as a component, or a function within the CCU-O fails the redundant unit will take over (leader - follower principle as shown on the picture below).
   The CCU-D is redundant ensuring that the diagnostic database is still accessible (from redundant device) if one fails.
   The CCU-S is redundant (warm standby), if one CCU-S enters fail safe state the redundant CCU-S take over the function (leader - follower principle as shown on the picture below).
9. Redundant TCN GW: The TCN GW is redundant (warm standby), if one TCN GW as a component, or a function within the TCN GW fails the redundant unit will take over (leader - follower principle as shown on the picture below).

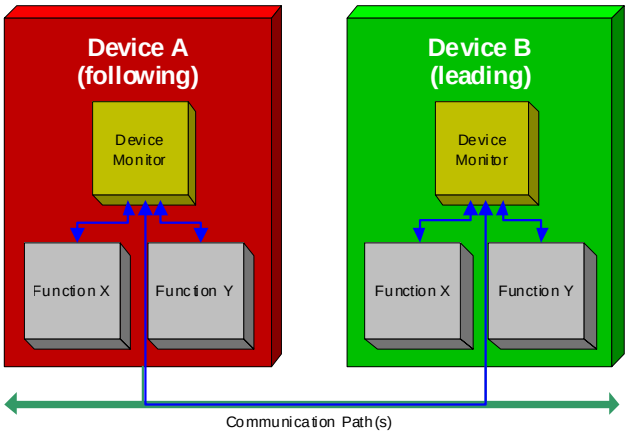


Figure: Leader / Follower Concept

The following failure modes and associated effects have been identified:

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Car	Item	Description
DM	CCU-O	CCU-O in other DM-car (if failing device was Leader) will take over as Leader.
DM	CCU-D	Diagnostics still available from CCU-D in other DM-car.
DM	CCU-S	CCU-S in other DM-car (if failing device was Leader) will take over as Leader.
DM	IDU primary	If primary IDU fails functionality will be activated on the back-up IDU. Switch fault will not affect the primary IDU as it is connected to two different switches.
DM	IDU back-up	Only active when primary IDU has failed.
M3	Crew IDU	Function lost. Switch fault will not affect the crew IDU as it is connected to two different switches.
Any	IP-RS	Loss of connected end devices. ECN communication re-routed, redundant end device activated (see CCU-O, CCU-D, IDU) were such exist.
Any	IP-TR	Loss of connected end devices. ECN communication re-routed, redundant end device activated (see CCU-O, CCU-D, IDU) were such exist. No impact of ETB communication.
DM	IP-TS	Loss of connected end devices. ECN communication re-routed, redundant end device activated (see CCU-O, CCU-D, IDU) were such exist. If this is the Leader the Follower IP-TS will take over, no impact of ETB communication.
DM	TCN GW	TCN GW in other DM-car takes over as Leader (if failing device was Leader). WTB communication not affected (except that a new train inauguration may take place).
Any	MIO	Input/output lost. Critical I/O accessible via redundant MIO device.
Any	BC	One MVB line (A or B) lost, communication still active via other MVB line.
Any	MOBAD Node ID	Device lost, redundant device activated.
Any	ECP-USB	Device lost, refer to description above (IP-RS, IP-TR, IP-TS, IDU)
	Communication cables	
ETB		One failing cable: no effect of function, redundant cable used instead.
ECN		One failing cable: no effect of function, traffic re-routed over the working ring links.
IP ED		ED communication lost.
WTB		One failing cable: no affect of function, redundant cable used instead.
MVB		**Scenarios similar to cable being disconnected:** Part of MVB lost (if no bus administrator on lost segment). **MVB terminator lost:** MVB segment between BC might be lost or disturbed. **Cable short circuit:** MVB segment between BC lost. CCU-O Leader / Follower switch logic set-up to minimize the consequence of any of the scenarios above.

This is covered by the requirements in the following sections and the behaviour of the used products.

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3 Functional requirements

3.1 Architecture

3.1.1 Function Design

The TCMS architecture drawing defines to structure of the TCMS used. In addition to that software architecture elements are used to split up different logical parts in the application software.

3.1.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.226
Description	The TCMS System shall provide a distributed communication network for train and vehicle level control and supervision.
Safety level	N/A

Master / Slave concept
The CCU-O application will support Slave and Master.
The Slave module shall execute in all CCU-Os.
The Master module shall execute in all CCU-Os which are in Master state.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.401	The CCU-Os in the consist with the active cab shall be set to Master.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.403	Master state shall be kept until cab in other consist is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The Master CCU-O, which is also Leader, shall activate the MVB source ports associated with the Master function.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.406	The Master CCU-O, which is also Leader, shall activate the IP redundancy groups associated with the Master function.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.407	An event shall be generated if more than one Master is active in the train.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Communication prototypes
The CCU-O shall implement the following communication prototypes:
- Master → Slave
- Slave → Slave

Note: Slave → Slave communication is intended also for communication from Slave to Master.

All Master-Slave, Slave-Slave communication will be sent on the train network (ETB, WTB), even if communication between Master and Slave module in same CCU-O.
Note: This is to ensure that the delay in communication is the same in all consists

ID	2F_09.04.Zefiro-Europe.TRS.227
Description	The TCMS system shall be built up by distributed intelligent computing units, touch screen displays, communication units and digital and analog I/O units.
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.411	The circuit diagram shall include the devices and connections as defined in the TCMS architecture, PGR_3VEVE10007THZ.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.3 Components

3.1.3.1 Function Design

3.1.3.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.255
Description	The CCU-O shall use a VCU-Compact type with MVB and serial channels.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.263
Description	The CCU-O shall contain the vehicle control application.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.256
Description	The CCU-D shall use a VCU-Compact type with MVB and serial channels.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.264
Description	The CCU-D shall contain the train diagnostic database.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.257
Description	A Gateway shall be used as a communication bridge between the local MVB and the train-wide WTB.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.376
Description	The CCU-S shall contain the safe vehicle control application.
Safety level	N/A

TCMS controllers

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.412	The compact units shall be installed in accordance with the PPC Installation Compact Units, 3EST000210-8999. And the project specific interface and installation manual 3EST000225-4873.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.413	The circuit diagram shall include the MVB device addresses for CCU-O, CCU-S, CCU-D and TCN GW as defined in the TCMS architecture, PGR_3VEVE10007THZ.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.89	The CCU-O shall contain the vehicle control application for functions allocated to TCMS software with safety integrity SIL0 or N/A.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.04.Zefiro-Europe.CONCEPT.470	The CCU-S shall contain the safe vehicle control application. Note: For functional requirements allocated to TCMS Software with SIL 1 or 2.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.95	The CCU-D shall contain the Train Diagnostic Database.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.414	The TCN GW shall be contain the GW-firmware application and configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.258
Description	I/O devices shall be used to interface a subset of connected systems and the vehicles hardware circuits.
Safety level	N/A

TCMS input/output units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.415	The MIO units shall be installed in accordance with the PPC MIO Product Description, 3EST000208-8462.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.416	The circuit diagram shall include the MVB device addresses for MIOs as defined in the TCMS architecture, PGR_3VEVE10007THZ.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.04.Zefiro-Europe.TRS.259
Description	An IDU shall be used for the interface to the driver.
Safety level	N/A

TCMS IDU units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.417	The IDU units shall be installed in accordance with the PPC HMI Handling Instruction, 3EST000219-8846.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.104	The IDU shall contain a GUI for train personnel.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.260
Description	IP-Ring Switches shall be used to build up the on board Ethernet Consist Network.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.270
Description	IP-Ring Switches shall be used to connect end devices which are using an IPT interface.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.261
Description	IP-Train Switches shall be used to connect the Ethernet Consist Network to the Ethernet Train Backbone (ETB).
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.262
Description	IP-Train Repeaters shall be used to ensure that the maximum allowed cable lengths of 90 meters between IP-Train switches/IP-Train Repeaters are followed for the Ethernet Train Backbone.
Safety level	N/A

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TCMS IPT units

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.116	IP switches, cables and connectors shall follow the PPC IPTCS Installation Manual 3EGM019001-0264 (old number 3EST000212-3090).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.418	The IP switches shall be configured with device names used on the network.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.419	The IP switch ports shall be configured to fixed 100bit full duplex.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.420	The IP switches shall be configured with multicast groups used on the network.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.268
Description	There shall be a MVB used for train operation control in each 4-car units.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.269
Description	There shall be a WTB used to interconnect the car MVB (for multiple operation of two 8-car trains).
Safety level	N/A

Mission critical communication

Mission critical communication will use MVB and WTB for communication in the train (except for needed communication with HMI and sub-systems).

3.1.4 Interfaces

3.1.4.1 Function Design

The main purpose of the TCMS is to integrate control and supervision of other onboard train systems.

3.1.4.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.271
Description	The following interfaces shall be used to interact with other systems on the train: – MVB – IP – IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.272
Description	The Propulsion (including DCU/L, DCU/M, DCU/F and DCU/A) shall use an MVB interface for operational control and supervision, IP for service access and diagnostics.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.273
Description	Brakes shall use MVB and IO
Safety level	N/A

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ID	2F_09.04.Zefiro-Europe.TRS.274
Description	Auxilliary Electric System shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.275
Description	Train Operation Control shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.276
Description	Passenger Doors shall use IP
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.277
Description	Passenger Information System and CCTV shall use IP
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.278
Description	ATP shall use MVB and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.279
Description	Bogie Monitoring shall use IP and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.280
Description	Air Supply shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.281
Description	Carbody Fittings shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.282
Description	Interior shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.283
Description	Toilet shall use IP
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.284
Description	Coupler shall use IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.285
Description	HVAC shall use IP
Safety level	N/A

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ID	2F_09.04.Zefiro-Europe.TRS.286
Description	Line Voltage shall use MVB and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.297
Description	The pantograph control shall use IP
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.287
Description	Battery shall use IP and IO
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.288
Description	Fire Detection shall use IP
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.289
Description	LIM shall use MVB
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.290
Description	Train Radio shall use MVB
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.182	The TCMS system shall implement the interfaces to the different systems connected to TCMS as defined in the system Interface Control Documents.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.2 Communication supervision

3.2.1 On Board system

3.2.1.1 Function Design

This section describes the communication supervision of the different TCMS components and of systems connected to TCMS.

3.2.1.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.207
Description	TCMS shall be used for train control, supervision and diagnostics.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.292
Description	TCMS communication between all devices in the TCMS System shall be supervised; An event shall be generated if communication with a specific device is not present.
Safety level	SIL 0

CSS based devices

Units with several cyclic tasks

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.185	Communication shall be supervised by lifesign per cyclic task level.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.382	Communication valid shall be set when all task lifesigns are updating.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.384	A unit communication error event shall be generated if not all lifesigns are updated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Thirdparty, CCU-S, Linux- and IO-devices

Units with only one lifesign

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.383	Communication shall be supervised by lifesign.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.385	Communication valid shall be set when lifesign is updating.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.424	A unit communication error event shall be generated if lifesign is not updated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Communication Network

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.426	MVB1 communication valid shall be set if communication valid to any of the devices connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.427	A MVB1 communication error event shall be set if none of the connected devices are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

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2F_09.04.Zefiro-Europe.CONCEPT.428	MVB2 valid shall be set if communication valid to any of the devices connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.429	A MVB2 communication event shall be set if none of the connected devices are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.471	WTB valid shall be set if communication valid to any of the devices connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.472	A WTB cable fault event shall be generated if GW indicate problem with WTB line A.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.473	A WTB cable fault event shall be generated if GW indicate problem with WTB line B.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.430	ECN communication valid shall be set if communication valid to any of the ECN switches.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.431	An ECN communication event shall be set if none of the ECN switches are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.432	ETB communication valid shall be set if communication valid to any of the ETB switches in other consist.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.433	An ETB communication event shall be set if none of the ETB switches in the other consist are reachable at the same time as WTB train configuration indicates multiple operation.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per switch

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.435	ECN switch valid shall be set if communication valid to any of the devices connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.436	An ECN switch communication event shall be set if none of connected end devices are reachable.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per ECN ring link

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.438	A ring link event shall be generated for each inter ECN switch link if the two adjacent ECN switches indicate ring fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Per ETB line

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.440	An ETB line error event shall be generated for each inter ETB switch line if the two adjacent Train Switches indicate that line is not trusted.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

Secondary faults

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.442	Secondary events due to lost communication shall be suppressed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.293
Description	Bus coupler shall be supervised and errors reported.
Safety level	SIL 0

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Bus coupler

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.421	Through each redundant pair of bus couplers only one MVB Line shall be connected per bus coupler. (I.e. Line A through one of the bus coupler, Line B through the other, see specification 3EGM032470).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.188	A Bus coupler error or MVB cable fault event shall be generated if MVB status word indicate problem with MVB line A.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.422	A Bus coupler error or MVB cable fault event shall be generated if MVB status word indicate problem with MVB line B.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

TCMS Status on HMI

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.519	TCMS shall communicates per car, the status of TCMS devices. In case of a failure is detected a red indication shall be set on IDU for the affected car.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

3.2.2 Redundancy

3.2.2.1 Function Design

To support different degraded scenarios redundancy is implemented to ensure that impact of single faults are mitigated.

3.2.2.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.209
Description	A single failure on the TCMS system shall not stop the train from continuing in operation. If one important unit (e.g. CCU-O, I/O, GW) fails a redundant unit shall handle the functionality of the failing unit.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.229
Description	The two CCU-Os shall implement the Leader Follower concept.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.236
Description	The two CCU-Os shall be connected to the same line redundant MVB.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.237
Description	The two CCU-Os shall be connected to the Ethernet Consist Network.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.244
Description	The MVB shall be implemented using line redundant communication cables.
Safety level	N/A

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MVB

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.197	The MVB1 and MVB2 shall be implemented using cables supporting MVB EMD line redundancy.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.443	The MVB1 and MVB2 shall be terminated in end devices. Note: Also the MVB ESD+ ports on the bus coupler shall be terminated to avoid EMC disturbances of the open connectors.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.479	The 8 car consist MVB including all CCU's shall work independently on the train communication (ETB, WTB).	Derived Requirement	4S_09.03.05.-.TCMS Software		

CCU-O Leader Follower

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.446	The two CCU-O in each consist shall implement the Leader / Follower concept as defined in PPC Product Catalogue 3EGM007200D2806.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.447	The CCU-O shall send lifesign on both MVB1 and MVB2 to the other CCU-O in the consist. (If either lifesign ok the counterpart is regarded as alive. Refer to LeadSel CI_V_T2 in PPC Application_Library 3EST000214-9687)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.448	The CCU-O shall send it redundancy state (Synchronization, Leader, Follower) on both MVB1 and MVB2 to the other CCU-O in the consist.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.449	Default Leader shall be CCU-O in DM1-car. (Refer to LeadSel CI_S_Decider in PPC Application_Library 3EST000214-9687)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.450	A CCU-O in Leader state shall be forced to Follower state if: • No Ethernet communication and other CCU-O have working Ethernet communication OR • No MVB1 communication and other CCU-O have working MVB1 communication (could also be the level of access to primary systems like propulsion/brake) OR • No MVB2 communication and other CCU-O have working MVB2 communication (could also be the level of access to primary systems like propulsion/brake) (Refer to LeadSel CI_S_Decider in PPC Application_Library 3EST000214-9687)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.451	The Leader CCU-O shall activate the MVB source ports associated with the Leader.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.452	The Leader CCU-O shall activate the IP redundancy groups associated with the Leader.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.231
Description	The Ethernet Consist Networks shall be implemented as a ring to provide redundant traffic pathways.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.233
Description	The Ethernet Train Bus (ETB) shall be implemented using redundant communication cables.
Safety level	N/A

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ID	2F_09.04.Zefiro-Europe.TRS.242
Description	The Wire Train Bus (WTB) shall be implemented using redundant communication cables.
Safety level	N/A

Train communication

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.203	The circuit diagram shall include the Ethernet Consist ring cables, the redundant Ethernet Train Backbone cables and the redundant WTB cables as defined in the TCMS architecture, PGR_3VEVE10007THZ.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.453	The redundant Ethernet Train Backbone and the redundant WTB shall be connected over the automatic coupler to other consist when running in multiple operation.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.04.Zefiro-Europe.CONCEPT.502	In case WTB communication is completely lost during multiple traction operation, TCMS shall command an urgency brake in the leading consist. UB can be released activating emergency operation mode or at standstill when the new train configuration is acknowledged	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

As soon as the train is stopped the driver will acknowledge new train configuration and can proceed running only using towing operation and electrically disconnecting the two trains by isolation of the automatic coupler electrical head. Running in double traction will be always possible using emergency operation mode.

ID	2F_09.04.Zefiro-Europe.TRS.303
Description	The focal point control (direction control of the network trafic in the ring) shall be located in the train switches
Safety level	N/A

Consist IPT communication

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.252	The train switch in DM1-car shall be configured as Leader and train switch in DM8-car shall be configured as Follower (ECN focal point).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.454	The Leader CCU-O shall be responsible for the switch control as defined in PPC Switch Firmware Users Manual, 3EGM019001-0055.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.378
Description	The two CCU-S in each consist shall implement the Leader Follower concept.
Safety level	N/A

CCU-S

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.476	The two CCU-S in each consist shall be configured to Leader / Follower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.379
Description	The MVB bus adminstrator shall be redundant.
Safety level	N/A

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ID	2F_09.04.Zefiro-Europe.TRS.380
Description	The usage of MVB bandwidth up to 95% shall be used for process data.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.444	Redundant MVB bus administrators (BA) shall be configured on MVB1 and MVB2.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.480	The redundant bus administrator for both MVB shall be implemented on the CCU-S	Derived Requirement	4S_09.03.05.-.TCMS Software		

ID	2F_09.04.Zefiro-Europe.TRS.235
Description	The two Gateways (GW) shall implement the Leader Follower concept.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.238
Description	The two Gateways (GW) shall be connected to the same line redundant MVB.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.239
Description	The two Gateways (GW) shall be connected to the two WTB cables.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.240
Description	The two Gateways (GW) shall be connected to the Ethernet Consist Network.
Safety level	N/A

GW Leader Follower

The two Gateways in each consist will implement the Leader / Follower concept as defined in PPC Product Catalogue 3EGM007200D2806.
The Gateways will send lifesign on both MVB1 and MVB2 to the other Gateway in the consist.
(If either lifesign ok the counterpart is regarded as alive.
Refer to CI_V_T2 in PPC Application_Library 3EST000214-9687)
The Gateway will send it redundancy state (Synchronization, Leader, Follower) on both MVB1 and MVB2 to the other Gateway in the consist.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.455	Default Leader shall be Gateway in DM1-car. (Refer to LeadSel CI_S_Decider in PPC Application_Library 3EST000214-9687)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

A Gateway in Leader state will be forced to Follower state if:

- No Ethernet communication and other Gateway have working Ethernet communication
  OR
- No MVB1 communication and other Gateway have working MVB1 communication
  OR
- No MVB2 communication and other Gateway have working MVB2 communication

(Refer to LeadSel CI_S_Decider in PPC Application_Library 3EST000214-9687)

The Leader Gateway will activate the MVB source ports associated with the Leader.
The Leader Gateway will activate the WTB source ports associated with the Leader.
The Leader Gateway will activate the IP redundancy groups associated with the Leader.

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ID	2F_09.04.Zefiro-Europe.TRS.246
Description	Critical functions to the train operation shall be implemented using redundant I/O.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.224	Redundant MIO units shall be connected to different battery busses and shall use separate miniature circuit breakers.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.2.3 Self diagnosis / Self reset

3.2.3.1 Function Design

In addition to the inter TCMS component supervision also component self diagnostics is used within the different controllers.

3.2.3.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.248
Description	During boot-up, the device shall perform self diagnosis of its main components of the on board system (boards, RAM, etc.), Power On Self Test - POST.
Safety level	SIL 0

The bootloader for CCU and GW will be configured to perform a power on self test as defined in PPC Product Catalogue 3EGM007200D2806.

ID	2F_09.04.Zefiro-Europe.TRS.249
Description	Moreover, all active system processes shall be cyclically checked on the basis of configuration, after minor failures and, if necessary, restarted.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.251
Description	In case of serious non-recoverable damage the complete system shall be reset and a controlled shutdown shall be performed.
Safety level	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.250
Description	The device shall not require manual activities to restore non-blocking hardware or software failures of its component.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.233	Mission critical cyclic tasks in the device shall be supervised by a software watchdog.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.460	The complete software in the device shall be supervised by a hardware watchdog (for the targets with such support).	Derived Requirement	4S_09.03.06.-.TCMS Hardware		SIL 0

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3.2.4 Time Control

3.2.4.1 Function Design

The TCMS system reads a defined master clock and distributes the time within the TCMS system and to systems connected to TCMS and needing synchronized time.

3.2.4.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.211
Description	TCMS system time shall be synchronized from the GPS.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.78	The Time synchronization shall be based on the time synchronization defined in PPC Product Catalogue 3EGM007200D2806.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.461	The two CCU-Os within a consist shall be configured as redundant consist time masters (Leader-Follower).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.462	The time master shall use the GPS reference time from the MCG in UTC format.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.463	An event shall be generated if the reference time differs more than 10 seconds from the time masters internal realtime clock,	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_09.04.Zefiro-Europe.TRS.296
Description	The TCMS units and other IP connected devices shall synchronise their system time against the TCMS on board time master.
Safety level	N/A

ID	2F_09.04.Zefiro-Europe.TRS.298
Description	The time used for event logging in the fire detection system shall be synchronised with the TCMS system time
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.239	The different units in the TCMS system shall use NTP to synchronize their system time against the consist time master.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.464	The consist time master shall distribute time zone and daylight saving information to the different units in the TCMS system using IP-PD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.465	The consist time master shall distribute time, time zone and daylight saving information to connected systems using IP-PD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.466	The consist time master shall distribute time, time zone and daylight saving information to connected systems using MVB-PD.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.2.5 SW Download mode

3.2.5.1 Functional Design

TCMS system will allow software download on third party systems only if particular precondition are verified to avoid possible damage and dangerous situation during train operation.
This mode can be enable by maintenance people using IDU.

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3.2.5.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.510	"Software download mode" can be enabled/disable by maintenance people by pressing a dedicated softkey on IDU (visible only if user is logged in as maintainer and only if all preconditions are true)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.511	Every time Software download mode is enabled an event shall be set visible on IDU. The event will remain active as long as the SW download mode is active	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.04.Zefiro-Europe.CONCEPT.513	Software donwload mode can be enabled only if ALL following precondition are set: – Train standstill – Parking brake applied – HV disabled (panto down and LCB open)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.515	If software download mode is enabled it shall not be possible to raise any pantograph, close any LCB and release parking brake	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.517	In case parking brake is released, any pantograph is raised, any LCB is closed or the train is moving, software download mode shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.04.Zefiro-Europe.CONCEPT.514	When the SW download mode is active, CCUO will propagate this information to all the third parties for which this information is needed in the ICDs	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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3.3 Train Integrity

3.3.1 Monitoring of Train Configuration

3.3.1.1 Function Design

The TCMS system includes the functionality to create a valid train configuration, i.e. which cars and consist exist in the train and the direction of travel for those cars. This information is e.g. needed to create the train overview on the TCMS IDU.

3.3.1.2 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.222
Description	The TCMS configuration shall be dynamic and support operation of 8 and 16 car trains.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.83	The train configuration shall be dynamic and support operation of 8 and 16 car trains.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.467	The train configuration from the GW shall always be confirmed by the driver on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.294
Description	The train configuration shall be available for other systems.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.242	The train configuration shall be made available for other systems.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.04.Zefiro-Europe.TRS.295
Description	A unique car id shall be configured for each car (UIC id), which correspondes to the id painted on the outside of the car.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.245	A unique car id shall be configured for each car (UIC id), which corresponds to the id painted on the outside of the car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.4 Fail Safe State Control

3.4.1 Function Design

If the HMI411 safe processor detects inconsistencies (screen supervision, watch dog or SDT error of input data) it enters the operational safe state (DOSS). From this state it is automatically recovered if the cause for DOSS is recovered. It can be recovered from a non operational safe state (NOSS) via power cycle. If CCU-O detects that HMI is in fail safe state an automatic power cycle shall be performed max two times, then the HMI shall be replaced.

To leave the safe state of MIO-S, CCU-S, GW-S also a power cycle is possible. This is not foreseen because:
- the MIO-S shall not be reset automatically, the important functions use (availability-) redundant
- the CCU-S and GW-S shall not be reset automatically, leader-follower redundancy shall cover the single errors for availability

3.4.2 Normal condition

In normal operation the HMI is working properly and TCMS is not commanding any power cycle. The power supply is connected through a relay normally closed contact.
In case an inconsistency is detected TCMS energizes the relays disconnecting HMI power supply.

3.4.3 Failure cases

1) Command relay is kept energized (IDU not supplied).
2) Command relay faulty (power cycle not possible).

3.4.4 Failure detection

1) Detected by the Driver during operation.
2) Detected by the Driver during operation.

3.4.5 Failure handling

1) Command relay shall be replaced during maintenance.
2) Replace IDU during operation. Command relay to be replaced during maintenance.

3.4.6 Requirements

ID	2F_09.04.Zefiro-Europe.TRS.250
Description	The device shall not require manual activities to restore non-blocking hardware or software failures of its component.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.486	If CCU-O detectes that IDU (HMI411) has entered the non-operational safe state and never before or once before after battery power had been applied, the IDU shall be restarted by cutting and re-applying the battery power via a MIO DO.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.04.Zefiro-Europe.CONCEPT.487	A MIO DO shall control a relay with a normally closed contact in the power supply chain of the IDU (HMI411). The power shall be cut via an active MIO DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.04.Zefiro-Europe.TRS.383
Description	Hardwired IDU restart push-buttons or MCB shall be accessible for restart of each IDU separately.
Safety level	

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Nr	Description	Type	Derived to	User Interface	SIL
2F_09.04.Zefiro-Europe.CONCEPT.499	The power supply of IDU (HMI411) shall be done via dedicated MCB to allow manual restart. One MCB per cab and one MCB in the crew compartment shall be foreseen.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.29			

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.30			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_09.04.Zefiro-Europe.CONCEPT.33			

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5 Revision Log

5.1 Revision log 3EST000225-7917_B

Revision:	**_B**		
Date:	**2018-05-16**		
Editor:	**G. Parisi**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_09.04.ISSUES.-.Zefiro-Europe.8	[2F_09.04.Zefiro-Europe.CONCEPT.479]:		see FCCB decision 8.4.2013 line 6 to relax the requirement 479. The local consist shall work independently on the WTB or ETB but the GW and the TS are needed also for consist local communication. The second sentence in the req has been removed
2F_09.04.ISSUES.-.Zefiro-Europe.9	[2F_09.04.Zefiro-Europe.CONCEPT.502]: [2F_09.04.Zefiro-Europe.CONCEPT.503]:	WTB communication loss	WTB communication loss will trigger Urgency Brake in the leading consist
2F_09.04.ISSUES.-.Zefiro-Europe.10	[2F_09.04.Zefiro-Europe.CONCEPT.509]: [2F_09.04.Zefiro-Europe.CONCEPT.510]: [2F_09.04.Zefiro-Europe.CONCEPT.511]: [2F_09.04.Zefiro-Europe.CONCEPT.513]: [2F_09.04.Zefiro-Europe.CONCEPT.514]: [2F_09.04.Zefiro-Europe.CONCEPT.515]: [2F_09.04.Zefiro-Europe.CONCEPT.517]:	Software Download Mode	Requirement added for SW download mode implementation. Precondition to be satisfied 1) train standstill 2) parking brake applied 3) HV disabled (panto down and LCB open)
2F_09.04.ISSUES.-.Zefiro-Europe.11	[2F_09.04.Zefiro-Europe.CONCEPT.519]:	TCMS status on IDU	After FCCB, has been decided that TCMS shall communicate to HMI, status of TCMS, per car. In case a failure occurs the releted icon in the Train status page shalle be set as red.
2F_09.04.ISSUES.-.Zefiro-Europe.12	[2F_09.04.Zefiro-Europe.CONCEPT.485]: [2F_09.04.Zefiro-Europe.CONCEPT.486]:	HMI power reset function moved to CCUO	HMI power reset is not needed to be executed by CCUS, so function has been moved to CCUO
2F_09.04.ISSUES.-.Zefiro-Europe.13	[2F_09.04.Zefiro-Europe.CONCEPT.515]: [2F_09.04.Zefiro-Europe.CONCEPT.517]:	SW download mode reset conditions update	A pantograph raising added as de-activate condition of SW download mode

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